Arkaprava Sinha | Graduate Research Assistant

University of North Carolina, Charlotte

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Education

Program	Institution	Year
PhD. Computer Science Supervisor: Dr Srijan Das	University of North Carolina, Charlotte *North Carolina, USA*	2023-Present
M.Sc. Data Science	Chennai Mathematical Institute *Chennai, India*	2018-2020
B.Sc. (Hons.) Statistics	Calcutta University *Kolkata, India*	2015-2018
Senior School (AISSCE)	Delhi Public School, Ruby Park *Kolkata, India*	2013-2015

Experience

University of North Carolina, Charlotte **Aug 2023 - Present**
Graduate Research Assistant

- Video Understanding in Long Untrimmed Videos for Actions of Daily Living
- Multimodal Learning for Action Recognition in Videos

Accenture AI **Nov 2021 - Feb 2023**
Data Scientist

- Generating Insights from thousands of Earning Transcripts of companies using Natural Language Processing
- Contribution to a Library of Algorithms for Accenture AI

Larsen and Toubro Infotech **July 2020 - Nov 2021**
Data Scientist

- Building Industry solutions using Image Processing and Computer Vision
- Reviewing Machine Learning and Deep Learning Products

Teradata India Pvt. Ltd. **May 2019 - July 2019**
Data Science Intern

- Implementation of Topic Modeling Frameworks for Teradata MLE
- Research on new approaches for Clickstream Analysis

Tata Steel India Ltd. **June 2018 - July 2018**
Intern

- Worked on the application of Data Analytics in HR
- Built a model to predict employee attrition based on Qualification, Experience, Age and Level of employees

Publications

1. **Arkaprava Sinha**, Dominick Reilly, Pu Wang, Francois Bremond, and Srijan Das, *"SKI Models: Skeleton Induced Vision-Language Embeddings for Understanding Activities of Daily Living"*, accepted at The 39th Annual AAAI Conference on Artificial Intelligence (AAAI 2025).
2. **Arkaprava Sinha**, Monish Soundar Raj, Ahmed Helmy, Pu Wang, and Srijan Das, *"MS-Temba: Multi-Scale Temporal Mamba for Efficient Temporal Action Detection"*, under review.
3. Dominick Reilly, Rajatsubhra Chakraborty, **Arkaprava Sinha**, Manish Kumar Govind, Pu Wang, Francois Bremond, and Srijan Das, *"LLAVIDAL: A Large Language Vision Model for Daily Activities of Living"*, accepted at The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025).
4. Rajatsubhra Chakraborty*, **Arkaprava Sinha***, Dominick Reilly*, Manish Kumar Govind, Pu Wang, Francois Bremond, and Srijan Das, *"LLAVIDAL: Benchmarking Large Language Vision Models for Daily Activities of Living"*, accepted at 2 NeurIPS Workshops 2024.
5. Mahmoud Ali, Di Yang, **Arkaprava Sinha**, Dominick Reilly, Srijan Das, Gianpiero Francesca, and Francois Bremond, *"Quo Vadis, Video Understanding with Vision-Language Foundation Models?"*, accepted at NeurIPS Workshop 2024.

Key Projects

1. *Highlight Detection and Video Summarization in Long Videos*
University of North Carolina, Charlotte

- Developing efficient spatio-temporal representations to summarize long videos while preserving key information.
- Designing algorithms to detect key frames that effectively capture crucial events and transitions in extended video sequences.

2. *Temporal Action Detection in Long Videos*
University of North Carolina, Charlotte

- Building models to identify and localize multiple overlapping actions occurring in long-duration videos
- Addressing challenges related to action co-occurrence, temporal boundaries, and efficient processing of lengthy video sequences.

3. *Multimodal Action Recognition for Activities of Daily Living*
University of North Carolina, Charlotte

- Integrating RGB video, textual descriptions, and skeleton-based motion data to enhance action recognition performance.
- Exploring fusion techniques to effectively combine multiple modalities for improved understanding of daily activities.

3. *Multimodal Large Language Models for Activities of Daily Living*
University of North Carolina, Charlotte

- Investigating how large language models can leverage multimodal inputs, including videos, text, and skeletal data, for contextual action understanding.

Technical Skills

- Programming Languages: Python, R, C++
- Packages: PyTorch, Tensorflow, OpenCV, ScikitLearn
- Softwares: Minitab, SPSS